INTRODUCTION TO FLUID MECHANICS

Department of Chemical Engineering

CHEM ENG 2O04
Course Notes

Velocity Flow Profiles – Laminar vs Turbulent

Fox & McDonald

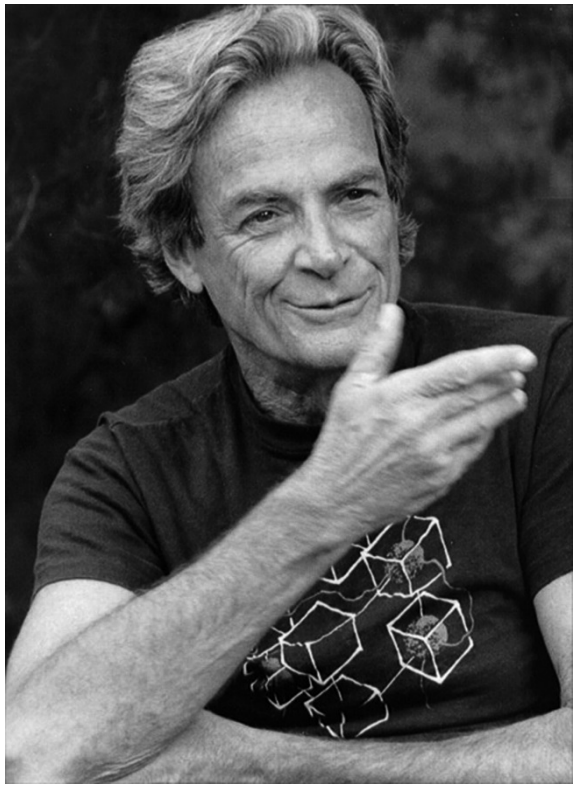
10th edition: 2.6, 7.3, 8.1, 8.5

9th edition: 2.6, 7.4, 8.1, 8.5

https://www.youtube.com/watch?v=9A-uUG0WR0w

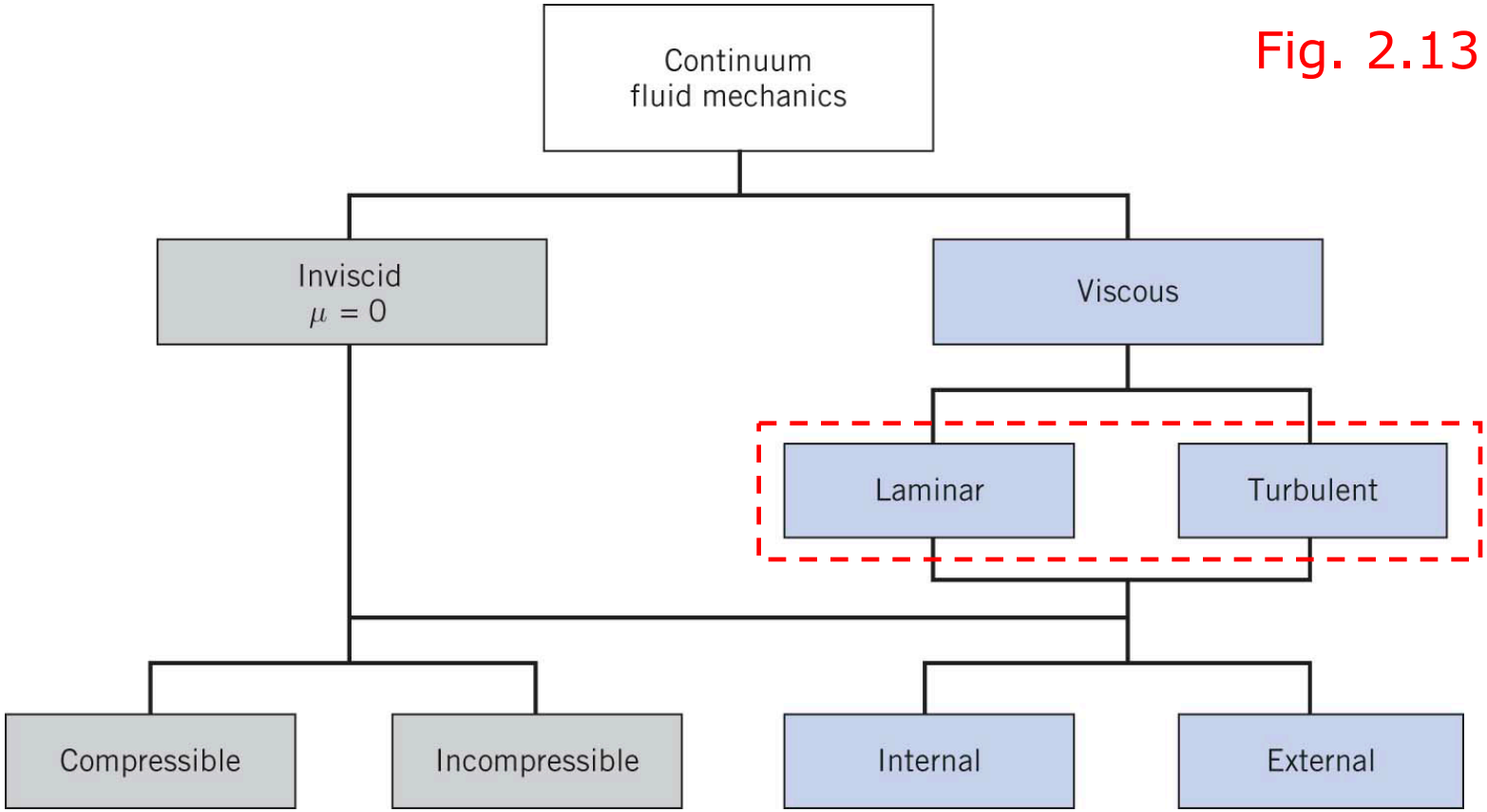
“I don't know what's the matter with people: they don't learn by understanding, they learn by some other way — by rote or something. Their knowledge is so fragile!”

Richard Feynman



Classification of Fluid Motions

Fig. 2.13



Laminar vs Turbulent Flow

- In 1883, Osborne Reynolds in his famous flow visualization experiments demonstrated the essential nature of laminar and turbulent flow

Reynolds Number

- The forces encountered in flowing fluids include those due to inertia, viscosity, pressure, gravity, surface tension, etc.
- The ratio of ANY two forces must be dimensionless

$$\text{Viscous force} \sim \tau A = \mu \frac{dV}{dy} A \propto \mu \frac{V}{L} L^2 = \mu V L$$

$$\text{Inertia force} \sim ma = \rho \cancel{V} \frac{dV}{dt} \propto \rho L^3 \frac{L/t}{t} = \rho \frac{L^2}{t^2} L^2 = \rho V^2 L^2$$

Characteristic Velocity

Characteristic Length

$$\frac{\text{INERTIA forces}}{\text{VISCOUS forces}} = \frac{\left(\rho V^2 L^2\right)}{\left(\mu V L\right)} = \frac{\rho V L}{\mu}$$

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$$\frac{\text{INERTIA forces}}{\text{VISCOUS forces}} = \frac{\left(\rho V^2 L^2\right)}{\left(\mu V L\right)} = \boxed{\frac{\rho V L}{\mu} = \text{Re}}$$

Eqn 7.11

Reynolds' number
Involved in virtually all types of flow problems

Figure 9.11 – Drag coefficient of a smooth sphere of diameter (D)

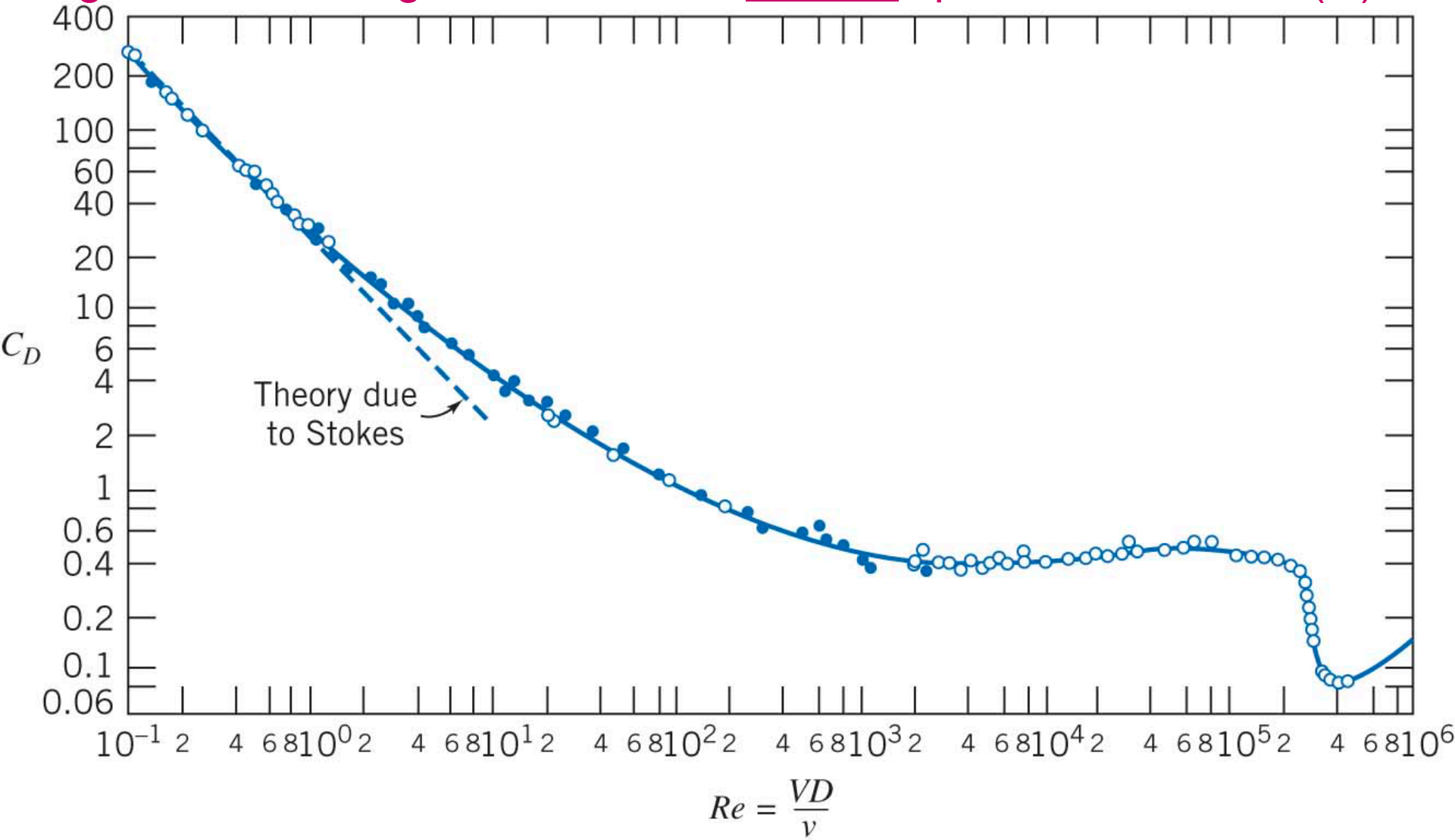

TABLE 2.4

Dimensions, Velocities, and Reynolds Numbers in the Canine Cardiovascular System

Vessel	Internal diameter (cm)	Length (cm)	Peak blood velocity (cm s^{-1})	Re_{peak}	Mean blood velocity (cm s^{-1})	Re_{mean}
Ascending Aorta	1.5	5	120	4,500	20	750
Descending Aorta	1.3	20	105	3,400	20	648
Abdominal Aorta	0.9	15	55	1,250	15	341
Femoral Artery	0.4	10	100	1,000	10	100
Arteriole	0.005	0.15	75	0.09	0.5–1	0.0006–0.0012
Capillary	0.0006	0.06	7	0.001	0.02–0.17	$2.86–24.3 \times 10^{-6}$
Venule	0.004	0.15	35	0.035	0.2–0.5	0.0002–0.0005
Inferior Vena Cava	1.0	30	25	700		
Main Pulmonary Artery	1.7	3.5	70	3,000	0.15	6.43

Source: Adapted from Ref. [19], with permission.

Laminar vs Turbulent Flow

- In 1883, Osborne Reynolds in his famous flow visualization experiments demonstrated the essential nature of laminar and turbulent flow
  - If Re is “small”, viscous effects will be dominant → LAMINAR flow
    - Fluid particles move in smooth layers (laminas)
    - NO lateral mixing

$$\boxed{\mathrm{Re} = \frac{\rho V L}{\mu}}$$

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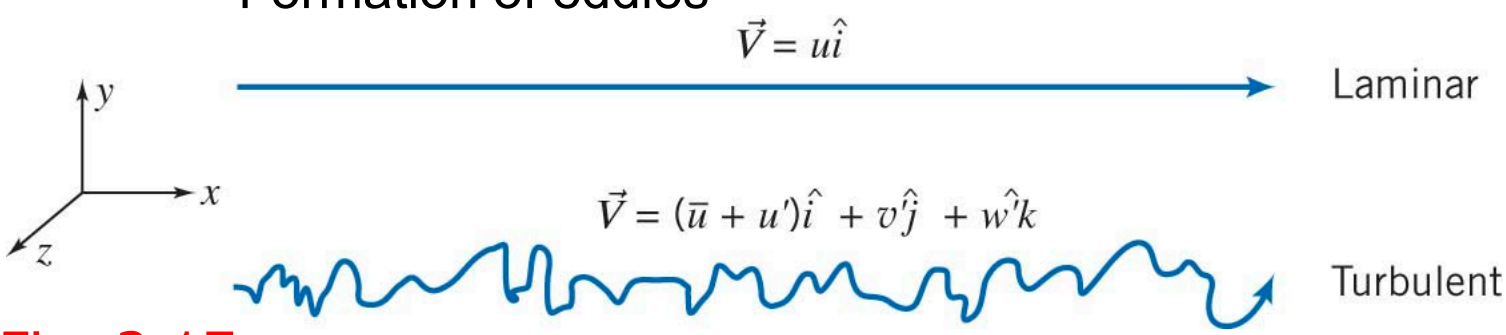
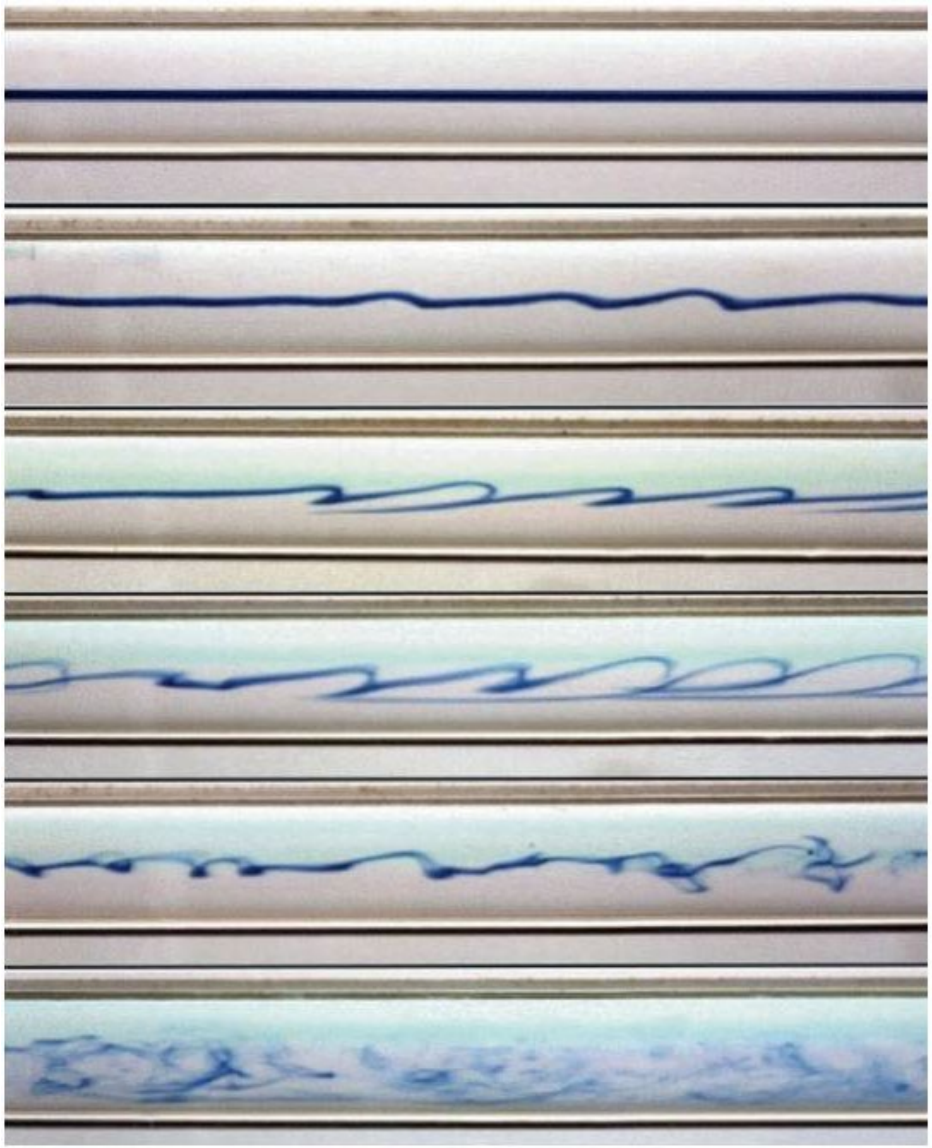
  - If Re is "small", viscous effects will be dominant → LAMINAR flow
    - Fluid particles move in smooth layers (laminas)
    - NO lateral mixing
  - If Re is "large", viscous effects are negligible → TURBULENT flow
    - Fluid particles rapidly mix as they move along due to chaotic/random 3D velocity fluctuations
    - Formation of eddies



$$\mathrm{Re} = \frac{\rho V L}{\mu}$$

Fig. 2.17

Reynolds Number for Pipe Flow



Very small Re

Very large Re

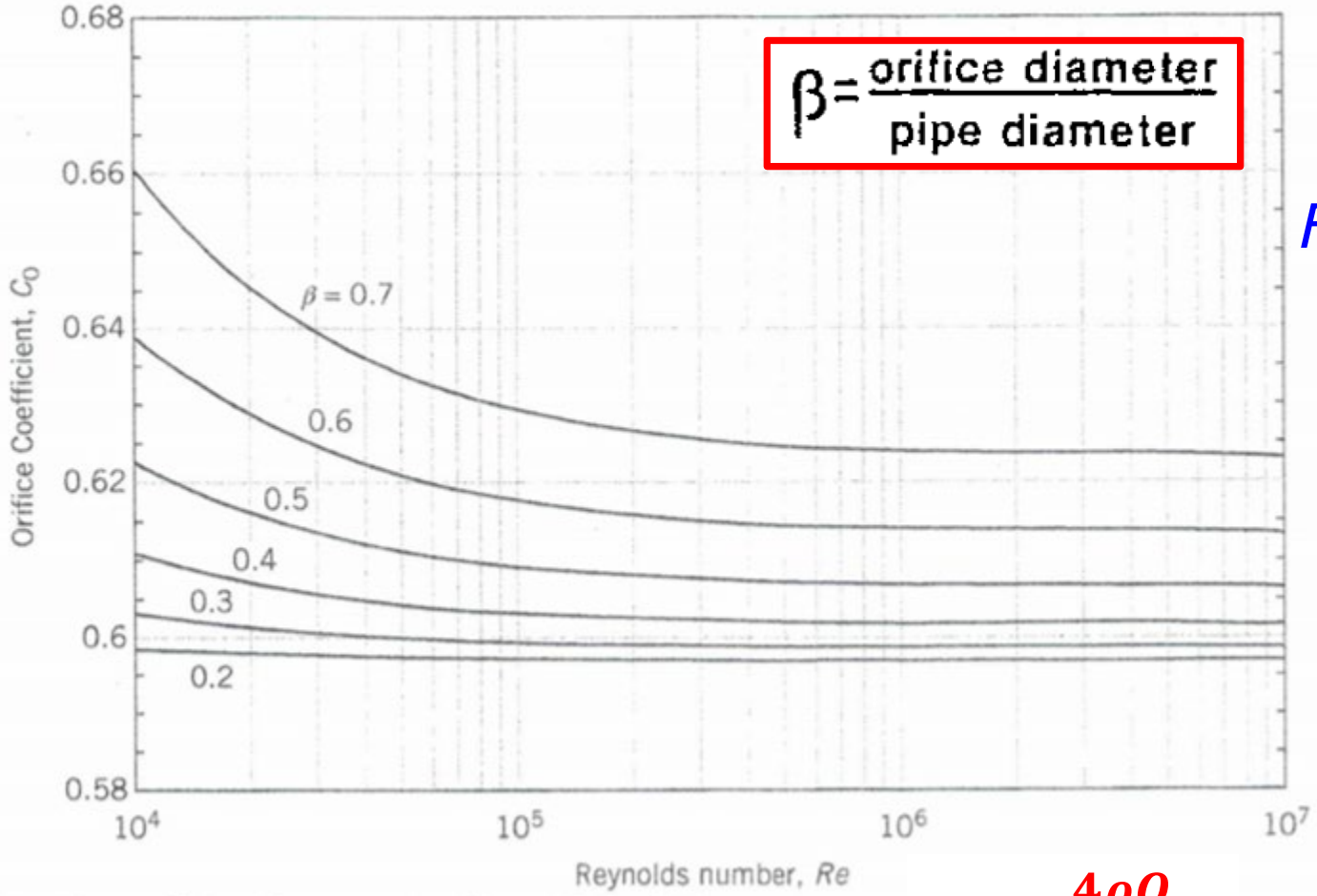
Reynolds Number for Pipe Flow

- For flow in a circular pipe:
  - The characteristic length L is the pipe diameter (D)
  - The characteristic velocity V is the average velocity (V_{avg} = Q/A) in the pipe

$$\boxed{\mathrm{Re} = \frac{\rho V_{avg} D}{\mu}}$$

$$\boxed{\mathrm{Re} = \frac{V_{avg} D}{\nu}}$$

Kinematic Viscosity



$$\beta = \frac{\text{orifice diameter}}{\text{pipe diameter}}$$

From PDF_04

Fig. 8.20 Flow coefficient for concentric orifices with D and $D/2$ taps.

$$Re = \frac{4\rho Q}{\pi \mu D_1}$$

ρ is the density of the fluid

Q is the volumetric flow rate of the fluid

μ is the viscosity of the fluid

D_1 is the diameter 'upstream' of the orifice meter

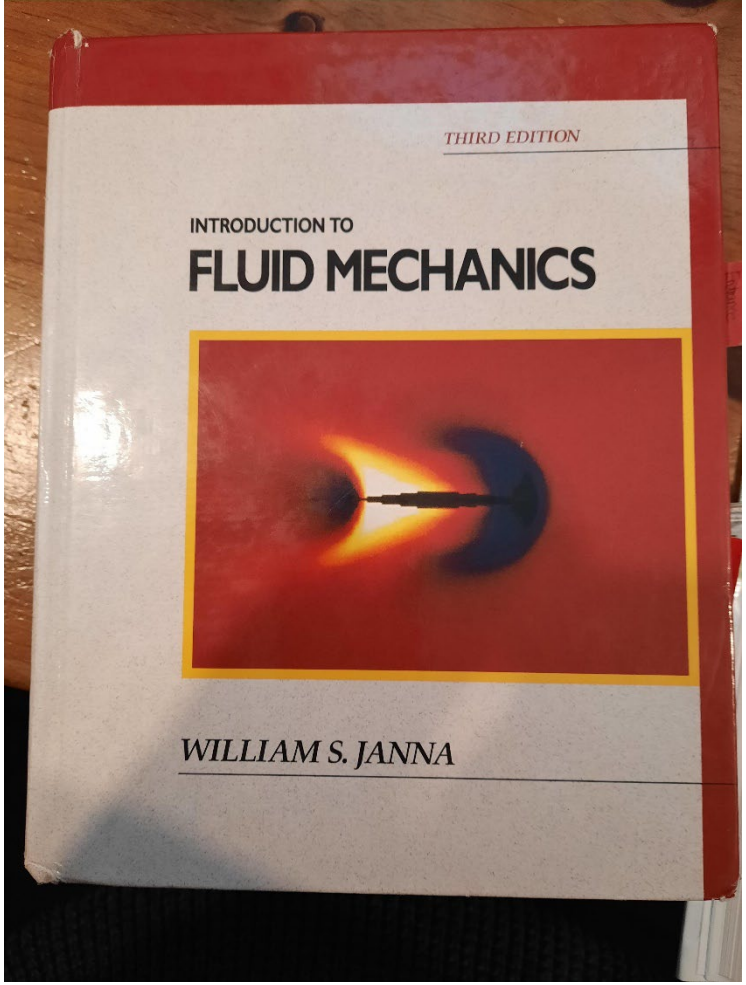
Reynolds Number for Pipe Flow

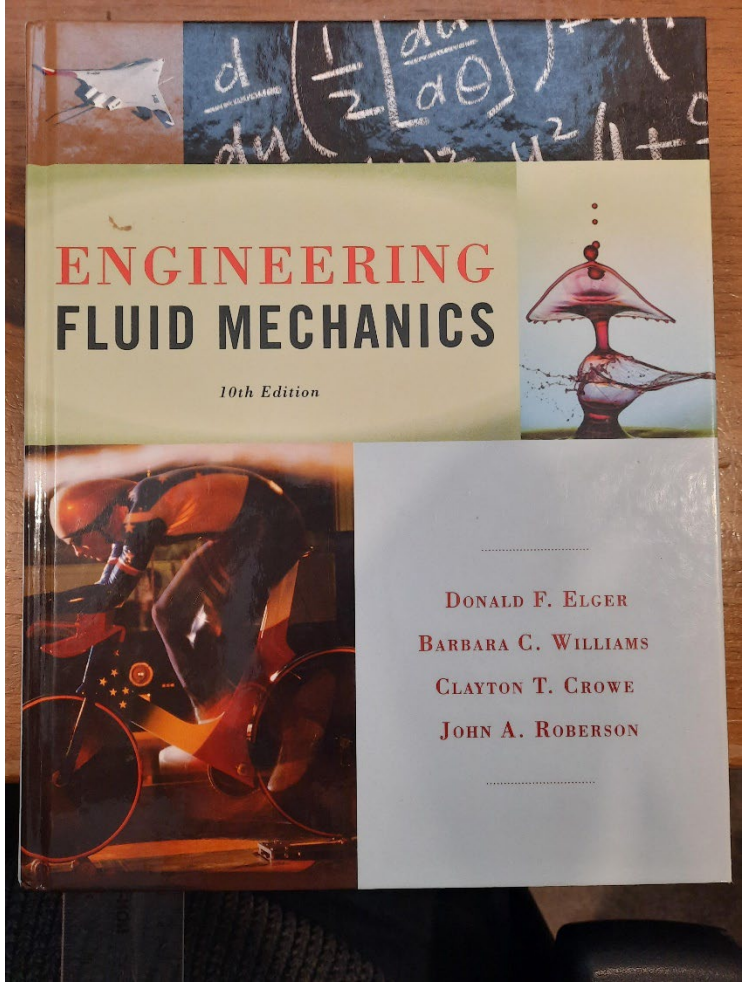
- For flow in a circular pipe:
  - The characteristic length L is the pipe diameter (D)
  - The characteristic velocity V is the average velocity (V_{avg} = Q/A) in the pipe

$$\boxed{\mathrm{Re} = \frac{\rho V_{avg} D}{\mu}}$$

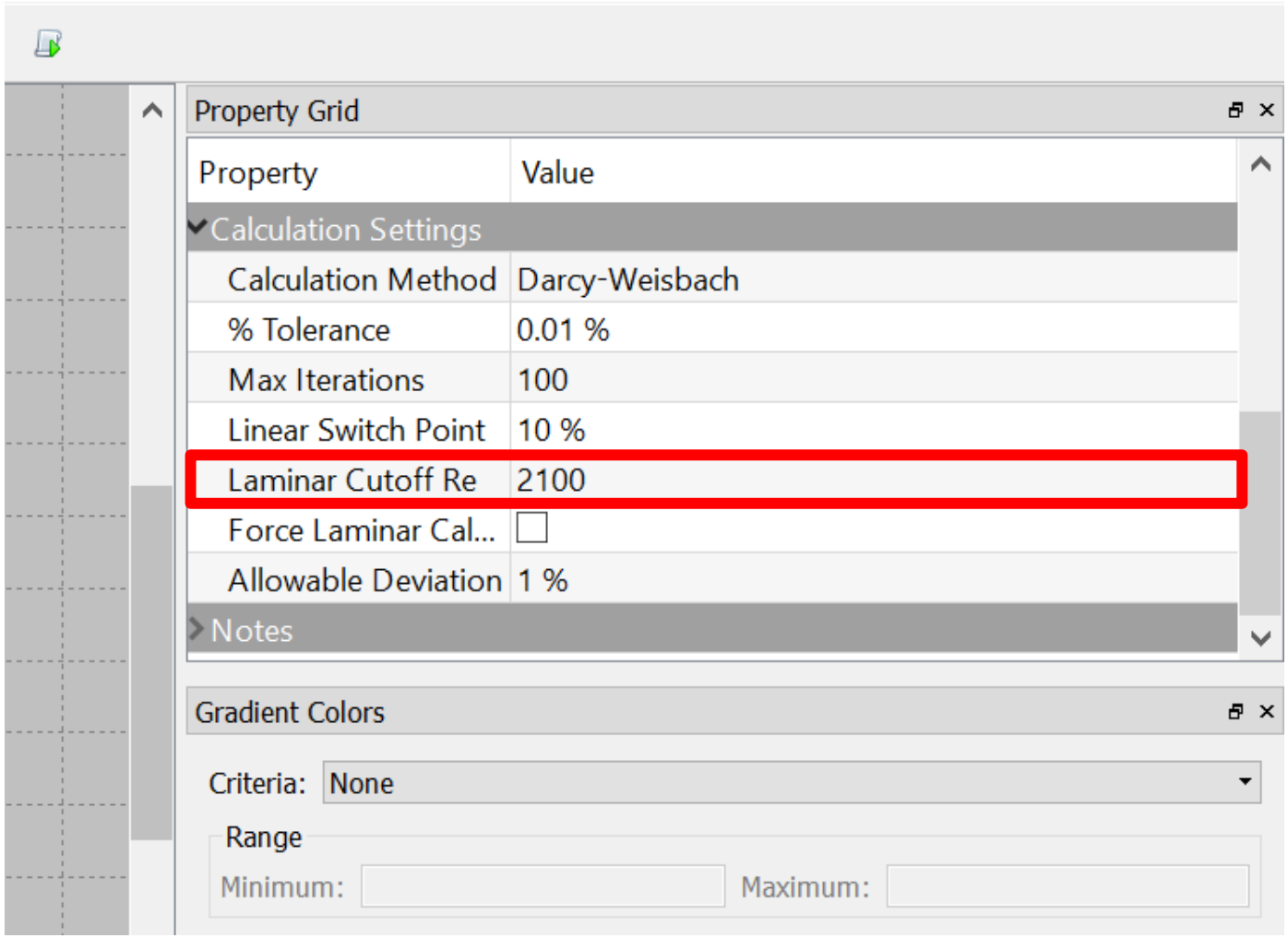
  - Laminar flow for Re < 2300
  - "Transition" flow for 2300 < Re < 4000
    - Combination of the two flow conditions
  - Turbulent flow for Re > 4000

According to your course textbook!

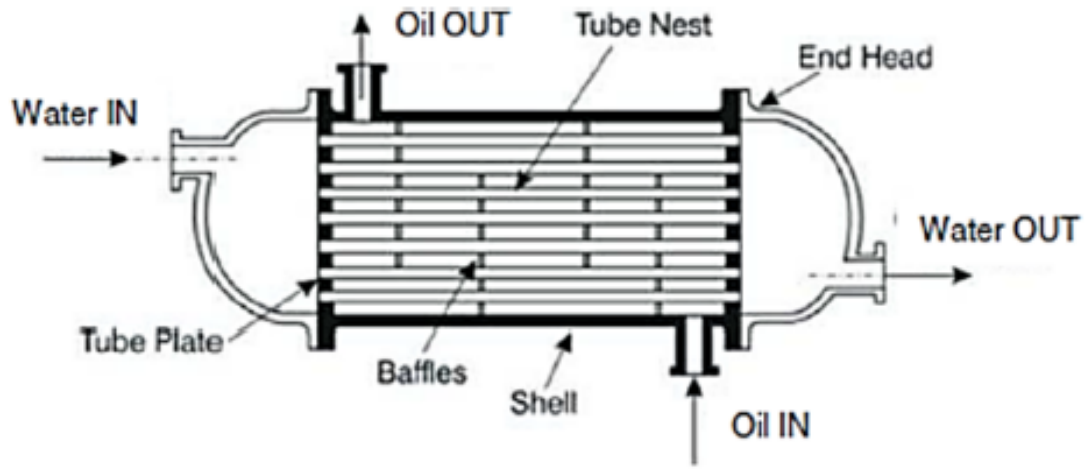

THIRD EDITION
INTRODUCTION TO
FLUID MECHANICS
WILLIAM S. JANNA


ENGINEERING
FLUID MECHANICS
10th Edition
DONALD F. ELGER
BARBARA C. WILLIAMS
CLAYTON T. CROWE
JOHN A. ROBERSON

PIPE-FLO default settings



In-class problem



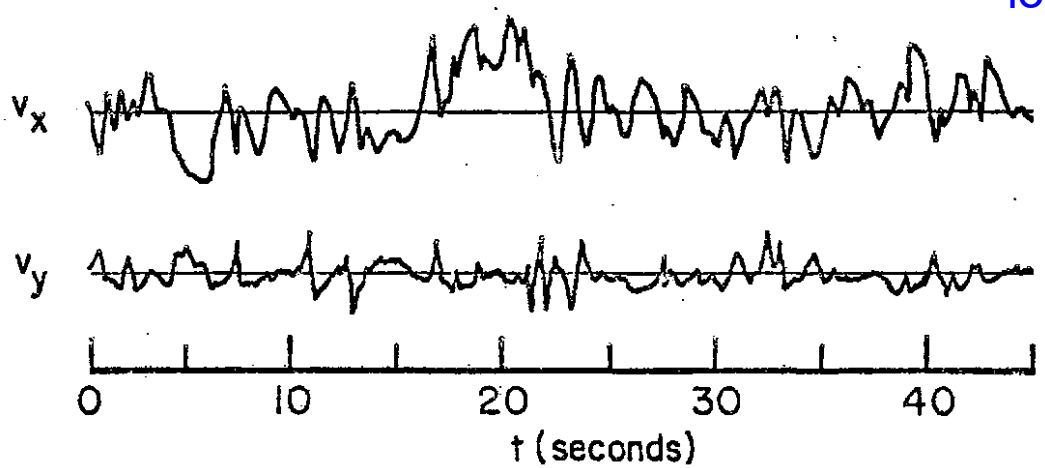
A shell-and-tube heat exchanger is a common unit operation in chemical plants. It consists of a bundle (also known as 'nest') of tubes inside a vessel (the shell). It is designed to transfer heat between two fluids. One fluid flows through the bundle of tubes while another fluid flows outside the tubes—but inside the shell. You will learn in ChE 3A04 that the *heat transfer coefficient* is dependent on the flow profiles of both fluid streams. As shown in the attached figure, a hot oil stream (shell side) is used to pre-heat a water stream (tube side) from 20°C (inlet temperature) to 50°C (outlet temperature). The inlet flow rate of water ($Q_{in} = 0.6$ m^3/min) is evenly-distributed across the bundle of fifteen tubes, each with inside diameter of 5 cm.

a. Calculate the Reynolds number (Re) of the inlet *and* outlet water streams <u>within each small tube</u>.

b. Comment on whether the inlet and outlet flows belong to the laminar, transitional, or turbulent regimes.

Turbulent Flow

- The flow field has enough kinetic energy to amplify small disturbances which are always present
- At any given point in a turbulent flow field, neither the velocity nor the pressure are constant



Instantaneous local velocity

Velocity fluctuation

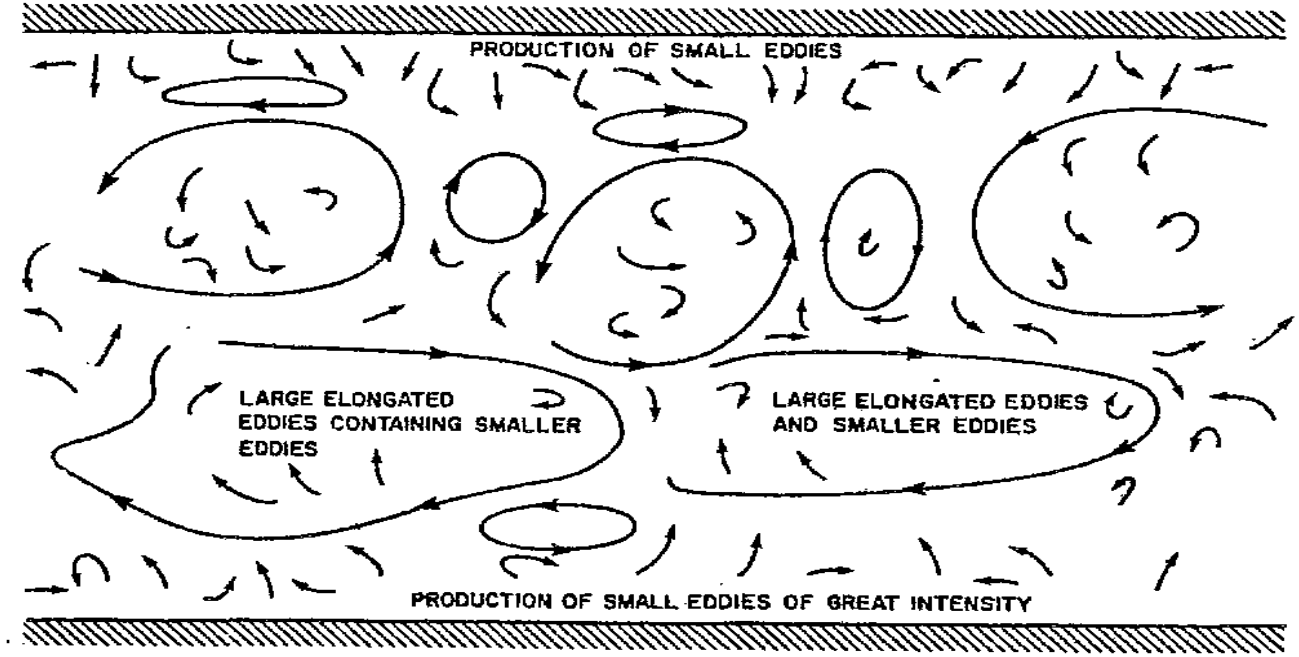
$$v_x = \bar{v}_x + v'_x$$

$$v_y = \bar{v}_y + v'_y$$

Time averaged local velocity

- Therefore, impossible to have complete description…but can distinguish an average velocity and pressure over a short time interval

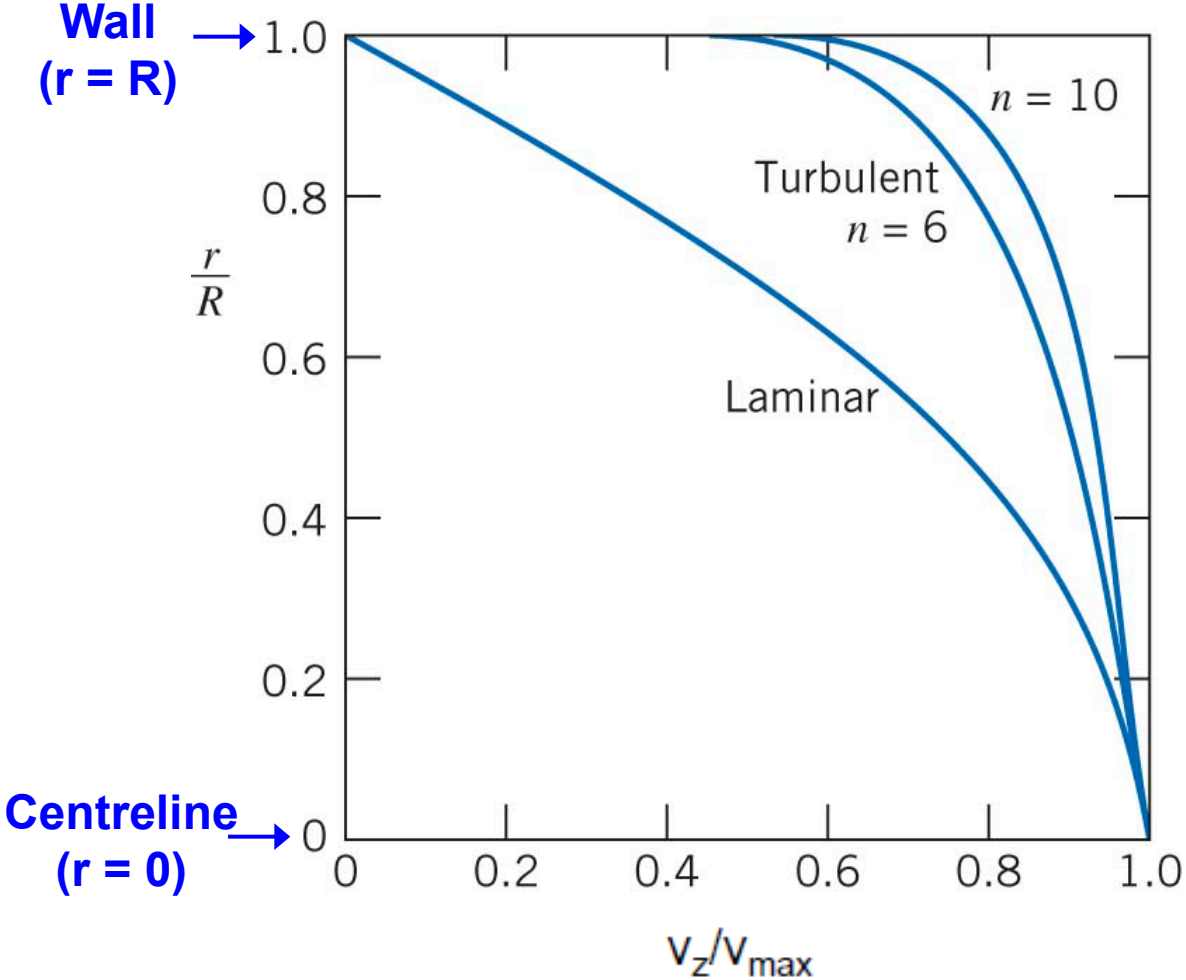
Eddy Structures in Turbulent Flow

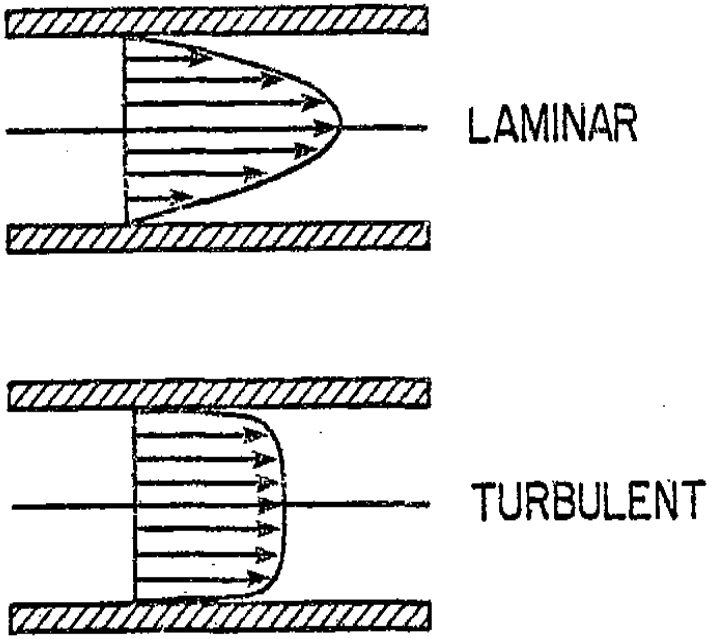


- Locally the velocity field exhibits gradients which correspond to recirculating flow and the formation of eddies
- Large velocity gradients near the wall break up the large eddies into smaller ones
- NO problem can be solved with the same rigorous mathematical analysis as was done for laminar flow
  - Rely on SEMI-EMPIRICAL models and analyses

Laminar vs Turbulent flow in a Pipe

Fig. 8.11





Laminar vs Turbulent flow in a Pipe

LAMINAR

$$v_z = v_{z,max}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$\frac{v_{z,avg}}{v_{z,max}} = \frac{1}{2}$$

Laminar vs Turbulent flow in a Pipe

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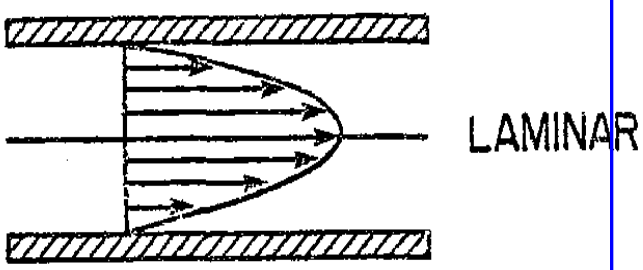
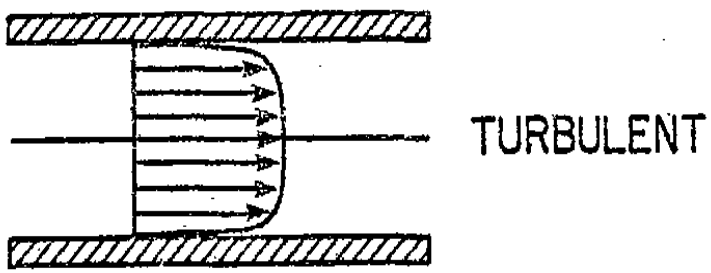


TURBULENT

$$\overline{v_z} = \overline{v_{z,max}}\left[1-\frac{r}{R}\right]^{1/n}$$

Eqn 8.22



Power-law Equation for Turbulent Flow in a Pipe

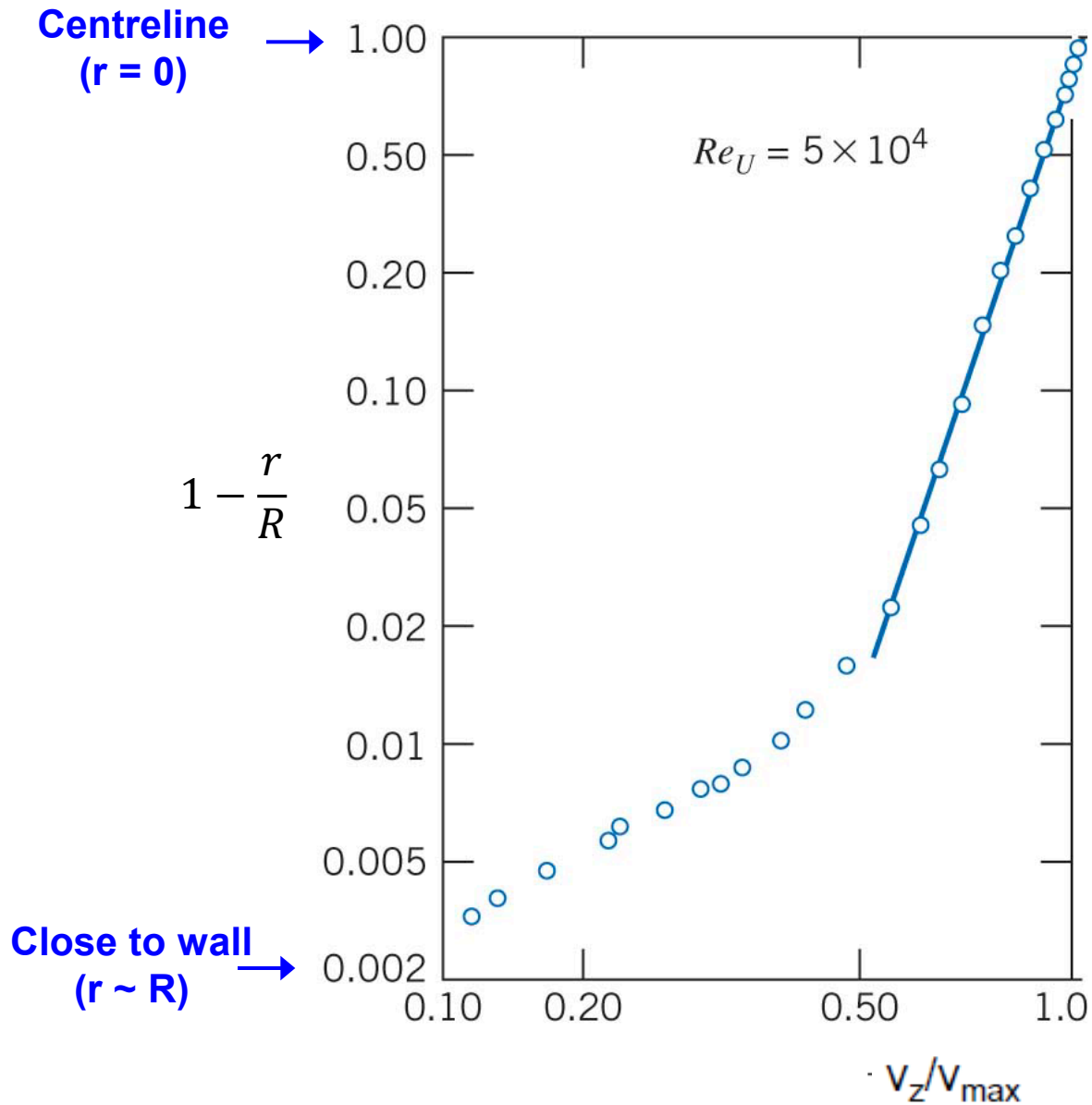


Fig. 8.10

The open symbols are data from an experiment.

The straight line is equation

$$\overline{v_z} = \overline{v_{z,max}} \left[1 - \frac{r}{R}\right]^{1/n}$$

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$$v_z = v_{z,max}\left[1-\left(\frac{r}{R}\right)^2\right]$$

$$\frac{v_{z,avg}}{v_{z,max}} = \frac{1}{2}$$

TURBULENT

$$\overline{v_z} = \overline{v_{z,max}}\left[1-\frac{r}{R}\right]^{1/n}$$ Eqn 8.22

The variation of power-law exponent 'n' with Reynolds number for fully developed flow in smooth pipes:

$$n = -1.7 + 1.8\log\mathrm{Re}$$ Eqn 8.23

Typically n = 7,
"one-seventh power profile"

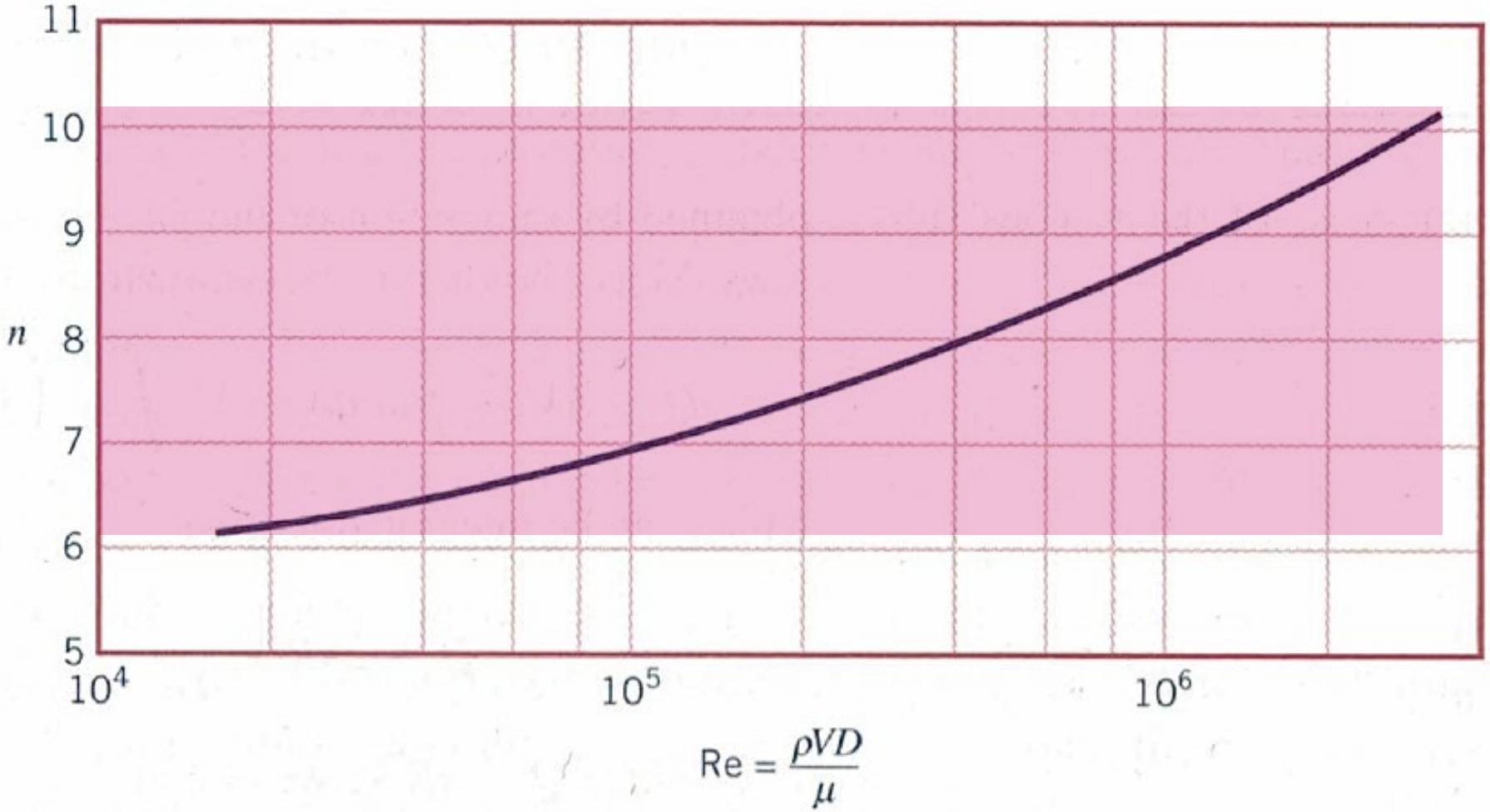
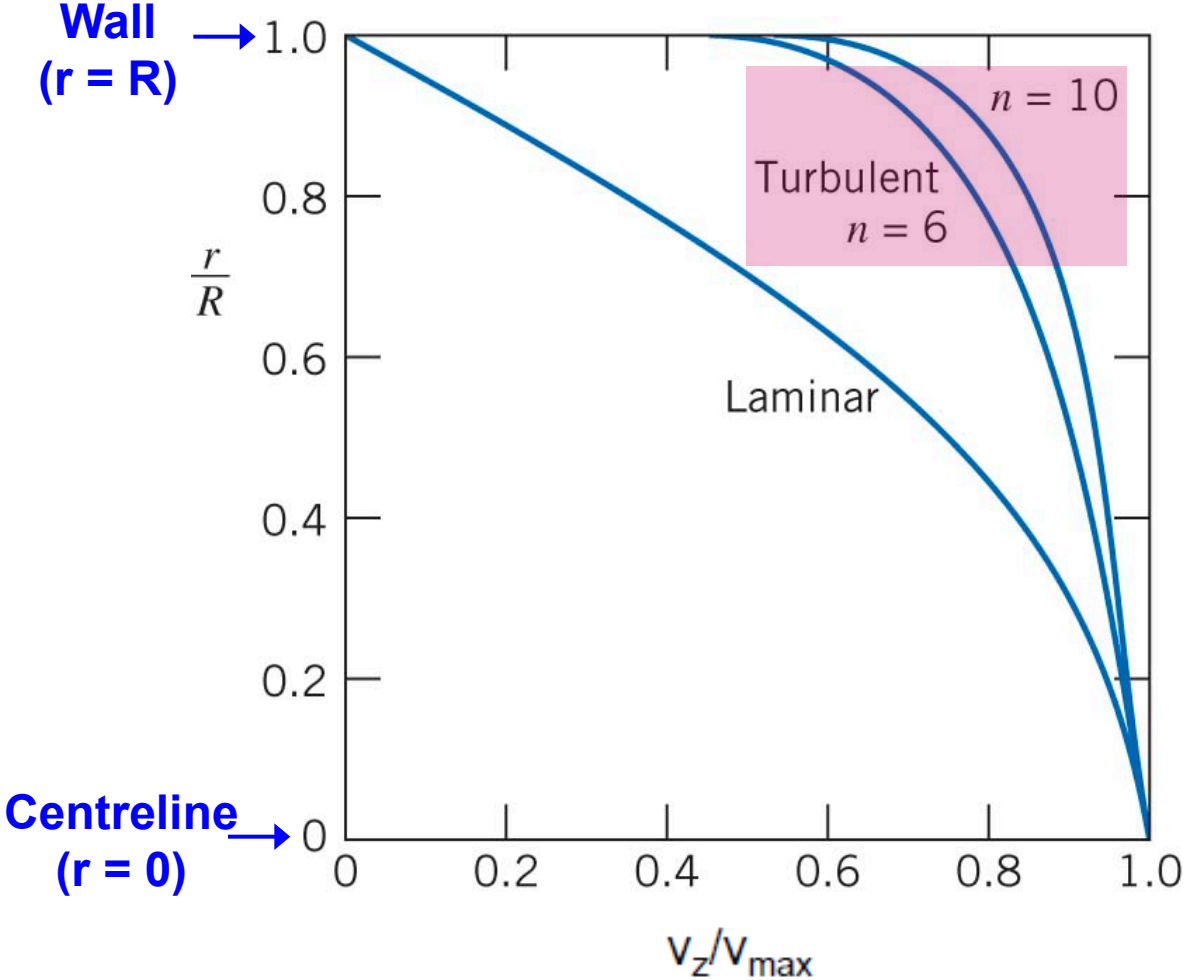
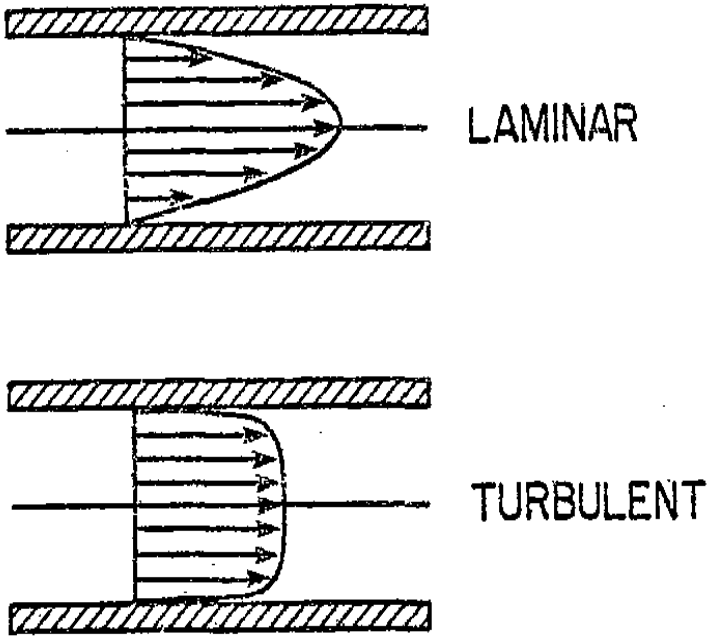


Figure 8.17 **Exponent, n, for power-law velocity profiles. (Adapted from Ref. 1.)**

Laminar vs Turbulent flow in a Pipe

Fig. 8.11





Laminar vs Turbulent flow in a Pipe

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TURBULENT

$$\overline{v_z} = \overline{v_{z,max}}\left[1-\frac{r}{R}\right]^{1/n}$$ Eqn 8.22

$$\text{for n} = 6, \frac{\overline{v_{z,avg}}}{\overline{v_{z,max}}} = 0.79$$

$$\text{for n} = 10, \frac{\overline{v_{z,avg}}}{\overline{v_{z,max}}} = 0.87$$

$$\text{in general}, \frac{\overline{v_{z,avg}}}{\overline{v_{z,max}}} \approx \frac{4}{5} = 0.8$$

Problems from textbook for self-study

10th Edition	9th Edition
In a food industry process, carbon tetrachloride at 20 ^{o}C flows through a tapered nozzle from an inlet diameter Din = 50 mm to an outlet diameter of Dout. The area varies linearly with distance along the nozzle, and the exit area is one-fifth of the inlet area; the nozzle length is 250 mm. The flow rate is Q = 2 Lmin. Does the flow exit the nozzle as turbulent flow? If so, at what point along the length of the nozzle does the flow become turbulent?	2.75
What is the Reynolds number of water at 20 ^{o}C flowing at 0.25 m/s through a 5mm diameter tube? If the pipe is now heated, a what mean water temperature will the flow transition to turbulence? Assume the velocity of the flow remains constant.	2.76
2.35	2.78
8.2	8.2
For flow in circular tubes, transition to turbulence usually occurs around Re~2300. Investigate the circumstances under which the flow of (a) standard air and (b) water at 15 oC become turbulent. On log-log graphs, plot: average velocity, volumetric flow rate, and mass flow rate at which turbulence first occurs as a function of the tube diameter.	8.3
8.21	8.54